

Hyeonjae Kim

Agency for Defense Development
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EDUCATION

Kyungpook National University (KNU)

Bachelor of Science in Physics

Overall GPA: 4.11/4.3 | Major GPA (71 credits): 4.19/4.3

Daegu, Republic of Korea

Mar. 2020 – Feb. 2024

PUBLICATIONS

- [4] **A zero-field optically pumped magnetometry platform for characterizing magnetic interference from electronic components used in portable atomic sensors**
Manuscript in preparation
H. Kim, S. Lee, S. H. Yim, T. Jeong, and Y. Lim
- [3] **Magnetic anomaly detection of an undersea tunnel using a drone-mounted all-optical pulsed atomic magnetometer**
Manuscript under review
H. Kim, S. H. Yim, S. Lee, E.-S. Bang, S. R. Yeo, Y. J. Kim, T. Jeong, and Y. Lim
- [2] **Novel sealing of glass-blown atomic gas cells using resistive heating wires**
Manuscript under review
S. H. Hong, Y. Lim, S. H. Yim, J. B. Nam, S. Lee, T. Jeong, and H. Kim
- [1] **Aging test of atomic vapor cell with Al_2O_3 wall coating on cubic glass**
Applied Optics **64**, 7932–7937 (2025). DOI: 10.1364/AO.572036
H. Kim, T. Jeong, S. H. Hong, J. B. Nam, S. Lee, Y. Lim, and S. H. Yim

SELECTED PRESENTATIONS

- [3] **Probing magnetic noise from portable sensor components using a zero-field optically pumped magnetometer**
Poster presentation; APS Division of Atomic, Molecular, and Optical Physics (DAMOP) Meeting (2026)
- [2] **Experimental systems for spectroscopy and zero-field OPM with atomic vapor cells**
Invited oral presentation; KISTI-SNU Joint Workshop (QNRC Joint Workshop) (2026)
- [1] **Lifetime extension of rubidium vapor cells by Al_2O_3 coating**
Poster presentation; APS Division of Atomic, Molecular, and Optical Physics (DAMOP) Meeting (2025)

RESEARCH EXPERIENCE

Agency for Defense Development (ADD)

Research Officer for National Defense (Military service)

Daejeon, Republic of Korea

Jun. 2024 – Present

Supervisors: Dr. Sin Hyuk Yim and Dr. Sangkyung Lee

Research in experimental atomic physics and quantum sensing

- Developed an automated absorption spectroscopy system with spectral modeling to characterize atomic vapor cells and contributed to their fabrication
- Built a zero-field optically pumped magnetometry platform to evaluate component-induced magnetic interference and provide design guidance for portable atomic sensors
- Deployed a drone-mounted atomic magnetometer for magnetic anomaly detection and established a quantitative framework for interpreting the acquired data

Kyungpook National University (KNU)	Daegu, Republic of Korea
<i>Undergraduate Researcher</i>	
• Novel Applied Nano Optics Lab (Advisor: Prof. Junyeob Yeo) Fabricated a flexible photoelectrochemical cell using laser processing • High Energy Physics Lab (Moon Lab) (Advisor: Prof. Chang-Sung Moon) Developed momentum reconstruction algorithms for silicon detector simulations	Jun. 2021 – Jun. 2022 Jun. 2020 – Feb. 2021

HONORS AND AWARDS

Physics Alumni Association Award and Scholarship (Outstanding Graduate)	Feb. 2024
Department of Physics Alumni Association, Kyungpook National University	
2nd Prize, 2023 MiliTECH Challenge	Dec. 2023
Hosted by KAIST; awarded by the Agency for Defense Development	
2nd Prize, 2022 MiliTECH Challenge	Dec. 2022
Hosted by KAIST; awarded by the Korea Military Academy	
3rd Prize, Academic Conference	Dec. 2021
College of Natural Sciences, Kyungpook National University	
Dean’s List	Nov. 2021, Apr. 2022, Apr. 2023
College of Natural Sciences, Kyungpook National University	

GRANTS AND SCHOLARSHIPS

National Science and Technology Scholarship	Mar. 2022 – Feb. 2024
Korea Student Aid Foundation (KOSAF)	
Academic Merit Scholarship (Hyoseok Scholarship)	Apr. 2022
Kyungpook National University Alumni Association	
Academic Excellence Scholarship	Aug. 2020, Feb. 2021, Aug. 2021
Kyungpook National University	

OTHER RESEARCH OUTPUTS

Buffer Gas Pressure Estimation for OPMs	Dec. 2024
Registered software; Korea Copyright Commission (C-2024-047156)	
Rubidium Vapor Cell Lifetime Monitoring and Analysis Tool	Dec. 2024
Registered software; Korea Copyright Commission (C-2024-054414)	

ADDITIONAL EXPERIENCE

Selected Participant, KFAS Study Abroad Workshop	Jul. 2026 – Jul. 2027
Natural Sciences; Korea Foundation for Advanced Studies (KFAS)	
Teaching Assistant, Classical Mechanics II (PHYS212)	Sep. 2023 – Dec. 2023
Teaching Assistant, Electromagnetism II (PHYS312)	Mar. 2023 – Jun. 2023
International Student Tutor, KNU	Feb. 2022 – May 2022

SKILLS

Programming: Python, Wolfram Language	Software: LabVIEW, SolidWorks, COMSOL Multiphysics
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